Enterprise Knowledge Workspace

Product Design Brief: AI Workflow for Knowledge Reuse and Decision Preparation

Version: Public English version v0.2

Purpose: A concise product case study for the English portfolio page.

Status: Product-grade prototype / demo-able MVP.

Boundary: This document does not claim production deployment. It contains no private files, internal company documents, or sensitive data paths.

This case study explores how scattered organizational knowledge can become a source-grounded, reviewable, and maintainable AI workflow.

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0. How to Read This Document

This document presents the product thinking behind the Enterprise Knowledge Workspace, a scoped AI product module designed for organizational knowledge reuse and decision preparation.

It is not a full engineering specification. It focuses on the product problem, user jobs, AI intervention points, product scope, workflow design, validation logic, trade-offs, and governance boundaries.

The core question is:

How can scattered organizational knowledge be turned into a traceable, reviewable, and continuously improving AI workflow, instead of remaining a one-off document search or chatbot experience?

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1. Product Overview

1.1 Background

In many organizations, valuable knowledge does not live in a single knowledge base. It is distributed across project documents, meeting notes, decision records, review materials, operating rules, retrospectives, customer feedback, and cross-functional communication.

As information accumulates, traditional folders, keyword search, and personal memory become increasingly insufficient:

- People remember that a similar issue was discussed before, but not where the source material lives.
- New team members spend significant time rebuilding context.
- Project owners need evidence for decisions, but the supporting materials are fragmented.
- Retrospectives are written once, then rarely become reusable organizational memory.
- Similar questions are repeatedly researched, summarized, and discussed.

The product is therefore not designed as a generic Q&A bot. It is designed as a knowledge workflow:

Retrieve relevant materials -> trace evidence -> synthesize conclusions -> support project retrospectives -> extract reusable learning -> improve through failure cases and evaluation.

1.2 Product Definition

The Enterprise Knowledge Workspace is an AI workspace module for knowledge reuse and decision preparation. It turns scattered documents and historical materials into a workflow that is:

- searchable,
- source-traceable,
- reviewable,
- retrospective-friendly,
- maintainable,
- and iteratively improvable.

Its value is not to let AI generate answers freely, but to help users complete knowledge work based on controlled sources and human review.

1.3 Current Status

The current version is a product-grade prototype / demo-able MVP. It validates the following capabilities:

- semantic retrieval over a defined source scope;
- source-grounded Q&A;
- a shift from single-turn Q&A to an Agent workflow;
- Router, Tool, Memory, Human Review, and failure case loops;
- evaluation and versioning mechanisms for continuous improvement;
- clear boundaries: no automatic write-back, no replacement of human judgment, and no claim of production deployment.

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2. Business Problem and User Jobs

2.1 Typical Knowledge Sources

Organizational knowledge may come from:

- project documents and phase deliverables;
- meeting notes and decision records;
- retrospective reports and lessons learned;
- business rules, process documents, and operating manuals;
- customer feedback, after-sales records, and quality issue records;
- review comments, internal discussions, and cross-functional collaboration materials;
- historical cases, competitor analysis, and management reports.

The challenge is not a lack of knowledge. The challenge is the lack of a sustainable product mechanism for reusing it.

2.2 Core Business Problems

Problem	What Happens	Impact
Scattered materials	Knowledge is spread across documents, notes, records, and discussions.	Users rely on memory, keywords, and manual follow-ups.
Reuse is difficult	Historical experience remains in individuals or isolated documents.	Teams repeat research and repeat mistakes.
Evidence is hard to trace	Conclusions are reused without clear source evidence.	Decision confidence and accountability are weakened.
Onboarding cost is high	New team members need to ask people and read many files.	Project continuity suffers.
Retrospectives do not become assets	Lessons learned are not indexed or continuously improved.	Organizational memory remains weak.

2.3 User Roles

User Role	Typical Need
Project owner / Product manager	Find historical decisions, project references, and supporting evidence.
New team member	Understand project context, key decisions, and relevant materials quickly.
Manager / Decision participant	Review the basis of conclusions, risks, and historical cases.
PMO / Knowledge management role	Maintain reusable knowledge and reduce repeated communication.

2.4 Core User Jobs

Users are not only searching for files. They are trying to complete knowledge tasks:

1. Find relevant materials.
2. Trace supporting evidence.
3. Synthesize conclusions across multiple sources.
4. Prepare project retrospectives.
5. Prepare decision or reporting materials.
6. Extract judgment references for future work.
7. Preserve reusable learning.
8. Continue the task through follow-up questions.

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3. Current Workflow and AI Intervention Points

3.1 Traditional Workflow

Figure 1: Wireframe workflow diagram

Traditional workflows do not fail because knowledge does not exist. They fail because knowledge is difficult to transform into reusable conclusions, decision evidence, and organizational memory.

3.2 AI Intervention Points

Workflow Step	Traditional Pain Point	AI Intervention	Product Value
Find materials	Users depend on keywords and memory.	Semantic retrieval and related passage recall.	Lower search cost.
Trace evidence	Users still need to read many documents manually.	Source citation and evidence extraction.	Make conclusions verifiable.
Synthesize conclusions	Cross-document synthesis is time-consuming.	Multi-document summarization and structured synthesis.	Improve preparation efficiency.
Prepare retrospectives	Retrospectives rely heavily on personal memory.	Retrospective templates and lesson extraction.	Turn experience into reusable knowledge.
Generate judgment references	Search results do not directly support next-step thinking.	Evidence-based judgment references.	Move from Q&A to knowledge workflow.
Improve quality	Errors are hard to track and learn from.	Failure case logs, eval sets, and version records.	Support continuous improvement.

3.3 Scoped AI Product Module

This project is not a generic Q&A bot. It is a scoped AI product module for knowledge reuse and decision preparation.

It includes the following modules:

Capability	Product Role
Knowledge retrieval	Find relevant materials and evidence.
Source traceability	Allow users to verify conclusions against original sources.
Knowledge-task Q&A	Help users complete knowledge preparation tasks, not just answer isolated questions.
Project retrospective support	Turn historical materials into structured retrospectives.
Memory mechanism	Preserve short-term task context and current goals.
Human review	Keep human judgment at key decision points.
Failure Case and eval loop	Convert failures into iteration inputs.
Version management	Make prompts, workflow changes, and evaluation results traceable.

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4. Key Product Decision: From RAG to a Trusted Context Layer

4.1 The Model Is Stronger, But the Business Problem Is Different

As foundation models become stronger, many low-risk, general-purpose, one-off text tasks can be completed directly by the model. A complex intermediate layer is not always necessary.

However, knowledge reuse is different. The key question is not whether the model can produce a fluent answer. The key questions are:

- Does the answer come from the specified source scope?
- Can the user trace it back to original evidence?
- Can the knowledge be updated as internal materials change?
- Can access and usage boundaries be controlled?
- Can human users review and approve important outputs?
- Can errors be logged, evaluated, and improved over time?

Therefore, the system needs a trusted context layer between the foundation model and internal knowledge assets.

4.2 RAG as Context Infrastructure

In this project, RAG is not positioned as the final product value or a technical selling point. It is treated as one implementation of a trusted context layer.

Its role is to provide:

- controlled source scope;
- source traceability;
- knowledge update mechanisms;
- reviewable evidence;
- organizational memory;
- and quality improvement through failure cases and evaluation.

> The value of the context layer is not to "make the model smarter." It is to make model outputs usable inside organizational knowledge workflows, where source, boundary, evidence, reviewability, evaluation, and rollback all matter.

4.3 Direct Model Call vs. Trusted Context Layer

Question	Direct Foundation Model Call	Trusted Context Layer
Internal knowledge	Cannot guarantee coverage of private or updated materials.	Uses specified internal materials.
Evidence	Source evidence is difficult to guarantee.	Provides traceable references.
Knowledge updates	Depends on model updates or manually supplied context.	Can update with the knowledge source.
Access boundaries	Hard to manage at document or domain level.	Can be designed around source scope and permissions.
Verification	Output may be plausible but hard to verify.	Users can return to original evidence.
Organizational memory	Hard to preserve as reusable assets.	Retrospectives, failure cases, and eval sets can accumulate.

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5. Solution Approach and Product Route

5.1 Why Not Just Use a Foundation Model?

A foundation model is useful for general explanation, writing, and synthesis. But it does not naturally know the organization's latest private materials, and it cannot consistently provide source-grounded evidence unless a source mechanism is designed.

Product decision:

> The foundation model can serve as the reasoning and language layer, but not as the source-of-truth layer for organizational knowledge.

5.2 Why Not Prioritize Fine-Tuning?

Fine-tuning can help with stable classification, output structure, tone, or repeated formatting. It does not solve the core needs of this case: source freshness, evidence traceability, and knowledge updates.

Product decision:

> Consider fine-tuning later for stable classification or formatting tasks. First, establish a trusted context layer for retrieval, evidence, and knowledge updates.

5.3 Product Route Decision Tree

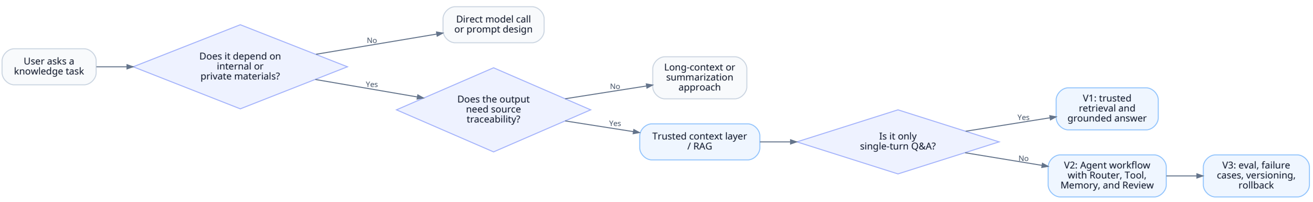

Figure 2: Wireframe workflow diagram

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6. Product Evolution: V1 / V2 / V3

The product evolution is not a story of increasing technical complexity. It is a story of turning organizational knowledge work into a more reliable product workflow.

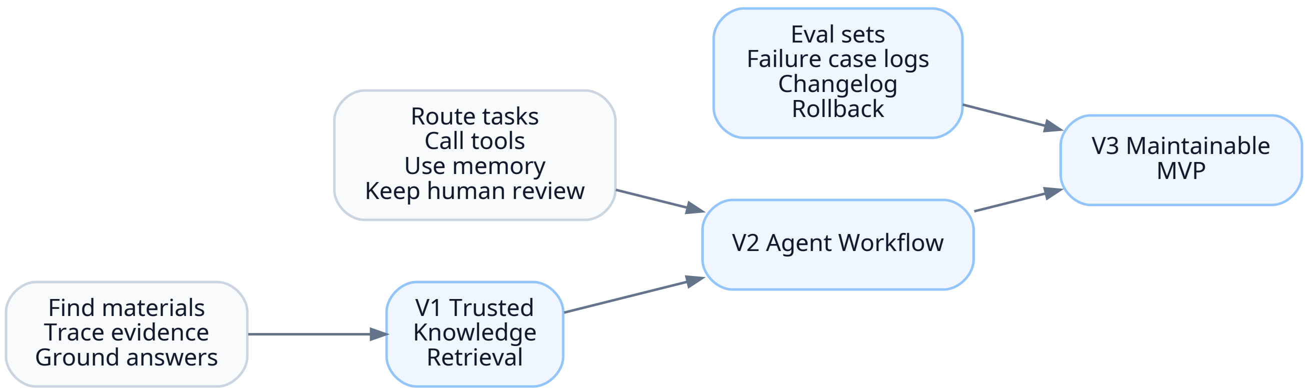

Figure 3: Wireframe workflow diagram

6.1 V1: Trusted Knowledge Retrieval

Key question:

> Why do we need a trusted context layer?

V1 focuses on making knowledge findable and answers verifiable:

- source configuration;
- document parsing and cleaning;
- text segmentation (chunking);
- vector representation (embedding);
- semantic retrieval;
- source-grounded answer generation;
- source display for human verification.

6.2 V2: Agent Workflow

Key question:

> Why is retrieval alone not enough?

Users do not only ask isolated questions. They often need to complete a sequence of knowledge tasks: find materials, trace evidence, synthesize conclusions, prepare a retrospective, and continue with follow-up questions.

V2 adds:

- Router: determine whether to answer directly, retrieve, or ask for clarification;
- Tool: package retrieval and synthesis capabilities as callable tools;
- Memory: preserve short-term task context;
- Agent Core: coordinate route, tool call, answer generation, and memory update;
- Human Review: keep human judgment at key output points;
- Execution Trace: show why the system retrieved, what it used, and how it answered.

6.3 V3: Maintainable MVP

Key question:

> Why is an Agent workflow still not enough?

Without evaluation, failure case tracking, prompt/workflow versioning, and rollback, an Agent workflow remains a demo rather than a maintainable product prototype.

V3 adds a lightweight workflow scaffolding layer:

Mechanism	Product Role
Memory Policy	Define what can be remembered and what should remain session-only.
Prompt / Workflow Versioning	Track prompt and workflow changes.
Changelog	Record meaningful product and workflow updates.
Decision Log	Preserve key product trade-offs and rationale.
Failure Case Log	Turn failures into iteration material.
Eval Set	Use fixed test cases for regression checks.
Rollback Note	Record failed versions and recovery paths.
Human Review Checkpoints	Define where human confirmation is required.

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7. Agent Workflow Design

7.1 Workflow Overview

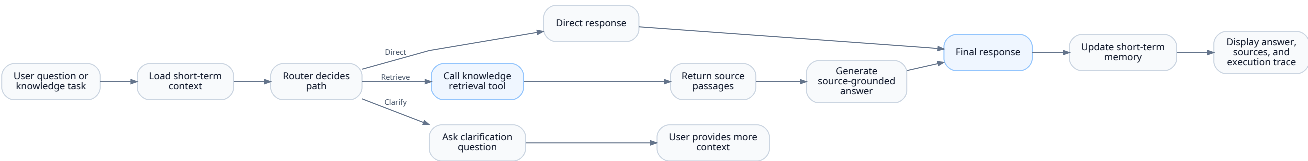

Figure 4: Wireframe workflow diagram

7.2 Core Components

Component	Product Meaning
Route Decision (Router)	Determines whether the user is searching, summarizing, preparing a retrospective, or asking an unclear question.
Tool Calling (Tool)	Turns retrieval, synthesis, and retrospective support into callable capabilities.
Trusted Context Layer	Provides traceable evidence and controlled source scope.
Memory Mechanism	Preserves short-term task context and current user goal.
Agent Core	Coordinates multi-step execution.
Execution Trace	Shows why retrieval happened, what sources were used, and how the answer was produced.
Human Review	Keeps human judgment before important outputs are accepted or reused.
Failure Case Review	Converts errors into future iteration material.

7.3 Safety Boundary

The prototype is read-only by default. It does not automatically write, delete, publish, or synchronize organizational knowledge.

Any future write-back, deletion, publication, or workflow-triggering action should require:

- preview;
- explicit human confirmation;
- version record;
- and rollback path.

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8. Validation and Failure Case Iteration

8.1 Validation Framework

The goal is not to prove that the AI can generate fluent text. The goal is to verify whether the workspace can support real knowledge work in a controlled and reviewable way.

Layer	Validation Focus
Retrieval layer	Whether the correct source appears in the retrieval results.
Answer layer	Whether the answer is grounded in the retrieved evidence.
Agent layer	Whether the Router chooses the right path and Memory does not pollute context.
Governance layer	Whether failure cases, eval sets, and version records support continuous improvement.

8.2 Key Metrics

Metric	Meaning
Source recall quality	Whether relevant sources are returned.
Citation support rate	Whether citations actually support the claims.
Route accuracy	Whether the system chooses direct, retrieve, or clarify correctly.
Tool success rate	Whether the retrieval tool returns usable evidence.
Memory accuracy	Whether follow-up questions preserve the right context.
Memory contamination rate	Whether topic changes incorrectly inherit old context.
Trace completeness	Whether route, tool call, sources, and answer are visible.
Safety gate effectiveness	Whether risky write/delete/publish actions require human confirmation.

8.3 Representative Failure Cases

Failure Case 1: Semantically Similar but Business-Irrelevant Retrieval

Item	Description
Symptom	A user asks about one historical project, but the system retrieves materials from another project with similar terminology.
Diagnosis	The issue is not answer generation. It is retrieval relevance: vector similarity does not always equal business relevance.
Fix	Add project metadata, source type, path filters, and query rewriting.
Product Principle	Retrieval must combine semantic similarity with business context and metadata.

Failure Case 2: Insufficient Evidence but Overconfident Answer

Item	Description
Symptom	Retrieved evidence is weak, but the model still generates a complete-looking answer.
Diagnosis	The answer layer does not properly handle insufficient context.
Fix	Require the answer to separate source-supported conclusions from uncertain points.
Product Principle	For knowledge work, trustworthiness matters more than fluency.

Failure Case 3: Router Misclassifies a Knowledge Task

Item	Description
Symptom	A user asks for a project retrospective, but the Router treats it as a generic writing task and does not retrieve sources.
Diagnosis	The control layer fails to recognize project-specific or source-dependent intent.
Fix	Add route rules for project, historical material, retrospective, and source-based questions.
Product Principle	In Agent workflows, the first product risk is often path selection, not model generation.

8.4 Failure Case Review Loop

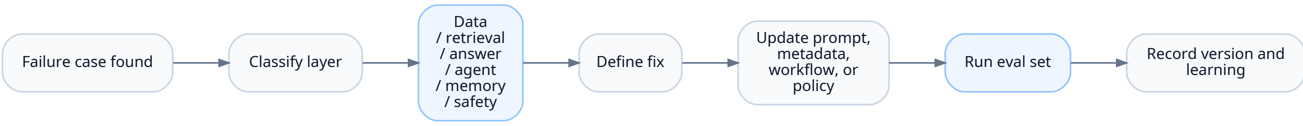

Figure 5: Wireframe workflow diagram

9. Human Review, Governance, and Scaling Considerations

9.1 Human-AI Collaboration Principles

Scenario	AI Can Do	Human Must Confirm
Knowledge retrieval	Recall sources and evidence passages.	Decide whether the evidence is valid.
Conclusion synthesis	Generate a structured draft.	Confirm conclusion accuracy.
Project retrospective	Extract events, issues, and lessons.	Approve final retrospective judgment.
Reference suggestion	Provide evidence-based judgment references.	Decide whether to act.
Knowledge write-back	Prepare draft content.	Confirm before any write-back or publication.

9.2 Governance Boundary

The system should not be positioned as a fully autonomous decision maker.

It should be positioned as:

- a source-grounded knowledge assistant;
- a retrospective preparation tool;
- a context layer for organizational knowledge;
- an AI workflow prototype with human review;
- and a maintainable product unit with eval and rollback mechanisms.

9.3 Scaling Considerations

To become a scalable product, the prototype would need additional capabilities:

Capability	Why It Matters
Role-based access control	Different users should see different knowledge scopes.
Multi-user workspace	Team knowledge needs collaboration and ownership.
System integration	Enterprise knowledge may live in document systems, project tools, CRM, or service records.
Audit logs	Users need to know who queried, generated, approved, or changed what.
Evaluation dashboard	Product owners need to monitor retrieval quality and answer reliability.
Sensitive data handling	Internal documents may contain confidential information.
Write-back workflow	Any update to knowledge assets should require preview, approval, and versioning.

10. Scope Boundary and Relationship with Other Modules

10.1 Current Scope Boundary

The current project is a product-grade prototype / demo-able MVP. It does not claim to be:

- a deployed production system;
- a complete permission and audit platform;

- a full task management system;
- an automatic write-back or publishing tool;
- a replacement for human judgment.

10.2 Boundary with the AI Requirement Flow Workspace

Module	Core Object	Key Question
Enterprise Knowledge Workspace	Knowledge assets, historical experience, decision evidence.	How can accumulated knowledge be retrieved, cited, reviewed, and preserved?
AI Requirement Flow Workspace	Current requirements, issues, risks, responsibility chain, workflow status.	How can new business signals be identified, confirmed, routed, tracked, and reviewed?

The Enterprise Knowledge Workspace may provide evidence-based references, but it should not expand into full requirement routing, ownership assignment, task status tracking, or operational workflow management. Those belong to the AI Requirement Flow Workspace.

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11. Public-Facing Summary

This project demonstrates how an AI product can move beyond a simple document Q&A tool. It reframes RAG as a trusted context layer, then builds an Agent workflow, human review, failure case iteration, and workflow scaffolding on top of it.

The key product value is not that the system can generate an answer. The key product value is that organizational knowledge work becomes:

- easier to search;
- easier to verify;
- easier to reuse;
- easier to review;
- easier to improve;
- and more suitable for future scalable productization.

Appendix A: Intentionally Omitted from the Public Version

The following content remains in the internal master document, but is intentionally excluded from this public English version:

- full implementation plans;
- detailed API schemas;
- raw test cases and full failure logs;
- private file paths or source names;
- interview scripts and internal portfolio notes;
- overly detailed engineering parameters.

This public version is designed to communicate product judgment, not to expose private implementation details.